

Learning Heritage by Touch: A Tangible Interactive Prototype for Changsha Kiln Mójíé Culture

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Abstract

Museum heritage learning is often “visible but not learnable” when artifacts are untouchable behind glass. We present a tangible prototype for learning the Changsha kiln Mójíé motif: a touchable ceramic relief block instrumented with pressure, temperature, and humidity sensing that maps touch/manipulation to three progressive content layers (form & craft, symbolic meaning, historical context). A web interface provides gradual disclosure and multimodal pacing cues to support a trajectory from perception to interpretation and contextual association. In a controlled two-condition study, the tangible condition outperformed a text-and-image baseline on a knowledge test and self-rated expression tasks, with the largest gains in form-detail identification and decorative-technique understanding. Interviews suggest tactile anchoring, contextual tactile cues, and iterative revisiting foster a reinforcing learning loop and emotional connection. We distill design implications for tangible heritage learning.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Interaction design.**

Keywords

Tangible Interaction, Cultural Heritage Learning, Haptic Feedback, Changsha Kiln, Mójíé Motif

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1 INTRODUCTION

Cultural heritage preserves historical memory and collective identity [22]. As museums adopt diverse formats and digital technologies, interactive and immersive installations have become common [18]. Yet two limitations persist. First, many systems increase “watchability” without improving “learnability,” and may distract

from artifacts and interpretation [4, 18]. Second, conservation keeps originals behind glass, limiting access to materiality, craftsmanship, and fine details, which weakens embodied engagement and emotional resonance [13, 21]. The challenge is to enable operable and perceptible learning while keeping originals fully protected.

Tangible interaction offers a path forward by coupling digital information with physical manipulation, reducing interface distance and supporting exploratory, embodied learning [5, 6, 11, 15]. Prior heritage systems show that touchable replicas and smart objects can trigger narratives, reconstructions, and multimodal cues to enhance engagement and understanding [2, 5, 17]. However, existing work is often case-driven: it reports engagement gains, but offers less explicit guidance on how learning is scaffolded over time, how content is structured for progressive understanding, and how outcomes are assessed in reusable ways [5]. Since cultural understanding is inherently layered—moving from observable form/material to symbolic meaning and then historical/value contexts [9, 14, 19]—tangible systems should map actions to a clear progressive pathway that guides visitors from “playing” to “learning” [10].

We study the Mójíé motif on Changsha kiln ceramics and present a tangible prototype that supports a “touch–manipulate–feedback–understand” chain. A touchable ceramic relief block serves as the input medium; interactions are mapped to three progressive layers: (1) form & craft details, (2) symbolic meaning, and (3) historical context. A web interface provides progressive disclosure and multimodal prompts to pace exploration and sustain engagement. We evaluate the system in a comparative study against a text-and-image baseline, reporting initial evidence and design insights for tangible heritage learning.

Our contributions are: (C1) a tangible prototype enabling touch-based learning without contacting originals; (C2) an explicit interaction–content mapping that links actions to learning across progressive interpretive layers; and (C3) initial empirical evidence and design implications for tangible learning in heritage settings.

2 DESIGN REQUIREMENTS

2.1 User Needs Survey: Touch, Depth, and Resonance

We conducted a survey, in-situ observation, and semi-structured interviews in the Changsha kiln gallery at Changsha Museum (see Appendix for photos), collecting 52 questionnaires (ages 17–52) and observing/interviewing 10 visitors. Participants valued the heritage’s historical (86%) and artistic significance (76%); 63% had prior exhibition experience. Three needs emerged: (1) Touchable

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Figure 1: The Mójie motif on Changsha kiln ceramics. Photo from the Changsha Kiln Research Group [3].

Interaction—92% wanted to touch and 80% expected haptic feedback, especially to feel form/material/texture via replicas (88%); (2) Deeper, Systematic Interpretation—79% wanted background, craft features, and cultural impact beyond surface descriptions; and (3) Immersion And Resonance—participants wanted richer interactivity (VR/AR 67%, physical 65%) and contextual cues (e.g., firing heat) to strengthen engagement and connection (82%). These findings motivate touch as a low-threshold entry and a progressively structured learning pathway (Table 2).

2.2 Field Study: “Touchable-to-Learnable” Potential of the Changsha Kiln Mójie Motif

We chose Changsha kiln ceramics for their distinctive craft features, cross-cultural exchange significance, and suitability for replica-based tactile learning. Fieldwork indicates the *Mójie* motif (Figure 1) is frequent and recognizable, with meanings shifting from an “exotic” fish-bodied beast to a localized symbol of protection and auspiciousness in Tang maritime exchange. As museum understanding often moves from form to symbolism to context [7], we structure content using Panofsky’s three-level framework [19]: form & craft → meaning → historical/transmission context (Table 1), enabling touch cues to support progressively deeper cultural understanding.

2.3 Requirements Summary: R1–R6 (Evidence & Implication)

Based on our user needs study (Section 2.1; 52 questionnaires, 10 observed/interviewed visitors) and field study of the Changsha kiln *Mójie* motif (Section 2.2), we derived six requirements for interaction–content mapping: **R1** Touch as entry (92% touch; 80% haptics; Section 2.1); **R2** Objects as cue carriers via durable replicas for pickup–touch–manipulate (88%; Section 2.1); **R3** Systematic depth beyond labels (79%; Section 2.1); **R4** Layered progression aligned with form & craft → meaning → context (Section 2.2; [7, 19]); **R5** Engagement/pacing through contextual resonance and progressive disclosure (82%; Section 2.1); **R6** Case fit: *Mójie* is frequent/recognizable and tied to maritime exchange, enabling stable tactile-to-cultural mappings (Section 2.2). Guided by **R1–R6**, our prototype uses a touchable *Mójie* replica to deliver progressively structured content.

3 PROTOTYPE

We present a tangible interactive prototype for learning the Changsha kiln *Mójie* motif. The system uses a touchable ceramic relief as

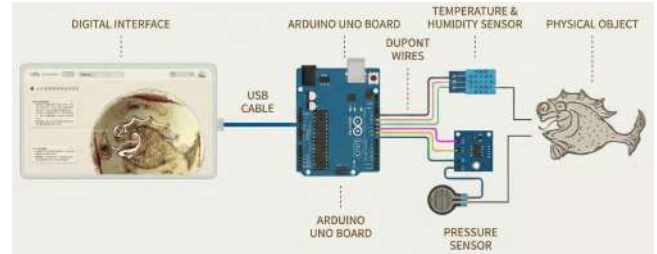


Figure 2: Sensor-to-module mapping and system architecture (pressure/temperature/humidity → events → progressive disclosure in the web UI).

the learning medium, turning touch and manipulation into “contact points” that trigger cultural content. Content is organized progressively—form & craft → symbolic meaning → historical/transmission context—to support meaning-making from perception to interpretation. The design operationalizes the Touch–Object–Culture requirements (Section 3): touch as entry (**R1**), a physical object as cue carrier (**R2**), structured and layered content (**R3–R4**), contextual prompts and feedback to sustain exploration (**R5**), and an interpretable, representative motif as the learning case (**R6**).

3.1 Prototype Overview and Design Logic

The prototype integrates (1) a touchable ceramic relief block, (2) pressure/temperature/humidity sensing, and (3) a web interface that provides guided content with multimodal feedback. Sensor signals are converted into interaction events that unlock content and prompts via progressive disclosure. Following Panofsky’s three-level progression, the system maps three cues to three modules: *press* → form & craft, *heat* → symbolic meaning, and *moisture* → historical/transmission context (Figure 2), guiding learners from perception to interpretation and contextual association.

3.2 Tangible Artifact Fabrication

The relief geometry was derived from representative *Mójie* features and slightly exaggerated to make key structures legible by touch (**R2**, **R6**). We fabricated a durable ceramic block suitable for repeated handling and integrated sensors/wiring on the back or inside to preserve a continuous exploration surface (Figure 3). Fabrication details are provided in the Appendix.

Table 1: Three-level structure of learning content for the Changsha kiln Mójíe motif.

Level	Learning Focus	Core Mójíe Points
Pre-iconographic	What is perceived (form & making)	Dragon-head fish-body; underglaze painting; layout and composition details.
Iconographic	What it means (symbolism)	Protective/water-deity links; auspicious mythical-beast sense; local fusion with dragon imagery.
Iconological	Why it matters (context)	Historical evolution; cultural exchange via the Silk Road; integration of Buddhist and indigenous Chinese motifs.

Table 2: User Needs – Pain Points – Design Implications for Tangible Heritage Learning

Dimension	User Needs	Current Pain Points	Implications for the Prototype
Experience	More interaction + contextual cues	Low interactivity; no touch; weak resonance	Haptics + multimodal cues; guided “explore–feedback–re-explore” loop
Object	Feel form/texture via touch	Originals untouchable; few touchable media; craft hard to perceive	Use touchable replicas/reconstructions; low-threshold actions
Content	In-depth, structured understanding	Static/fragmented info; meanings hard to reach	Progressive structure: form & craft → symbolism → context

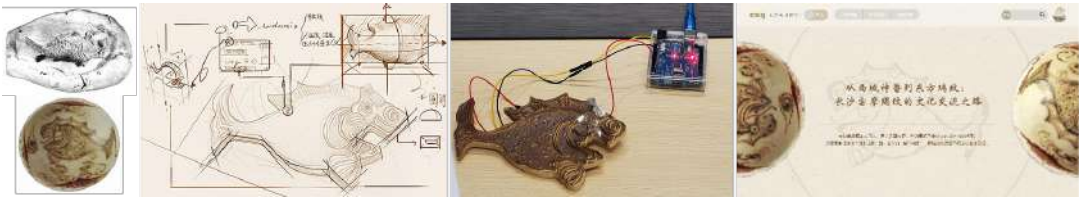


Figure 3: Reference motif examples (a), design sketches (b), finished touchable relief block (c), and the home interface (d).



Figure 4: From left to right: module UIs for (a) Form and Craft, (b) Symbolic Meaning, and (c) Historical and Transmission Context.

3.3 Module Design & Implementation

The interface includes a Home page (Figure 3d) and three modules (Figure 4) aligned with the layered learning structure: form & craft → meaning → context. Across modules, progressive disclosure and multimodal cues pace exploration and encourage revisiting. Arduino-sampled sensor data are streamed to a Node.js backend and pushed to the web UI in real time via WebSocket. The UI thresholds the signals into events to trigger modules/content and uses debounce plus short cooldowns to avoid false repeats and flicker (R3–R5).

Module 1: Form & Craft Details (Press → Form)

Goal. Help learners identify key form and craft details of the Mójíe motif and link touched location → visual detail → terminology, corresponding to Panofsky’s pre-iconographic level (what is seen/touched).

Interaction. Two-stage pacing: light touch/press highlights the touched region and shows 1–2 key points; firm/sustained press reveals deeper content (detail comparisons, composition, underglaze technique), moving from recognition to understanding.

Tactile cues. Relief geometry externalizes features (e.g., dragon head/fish body contour, fin/tail textures) so exploration can start without long reading.

Implementation. A pressure sensor (Arduino) triggers events when exceeding a threshold (600) with debounce to enter the module or unlock content.

Module 2: Symbolic Meaning (Heat → Meaning)

Goal. Build on form recognition to explain symbolism and meaning shifts (what it symbolizes).

Interaction. Temperature cues simulate firing: placing the block on a heated surface gradually raises temperature, triggering meaning cards (protection/auspiciousness) with kiln-fire sound to evoke context and resonance.

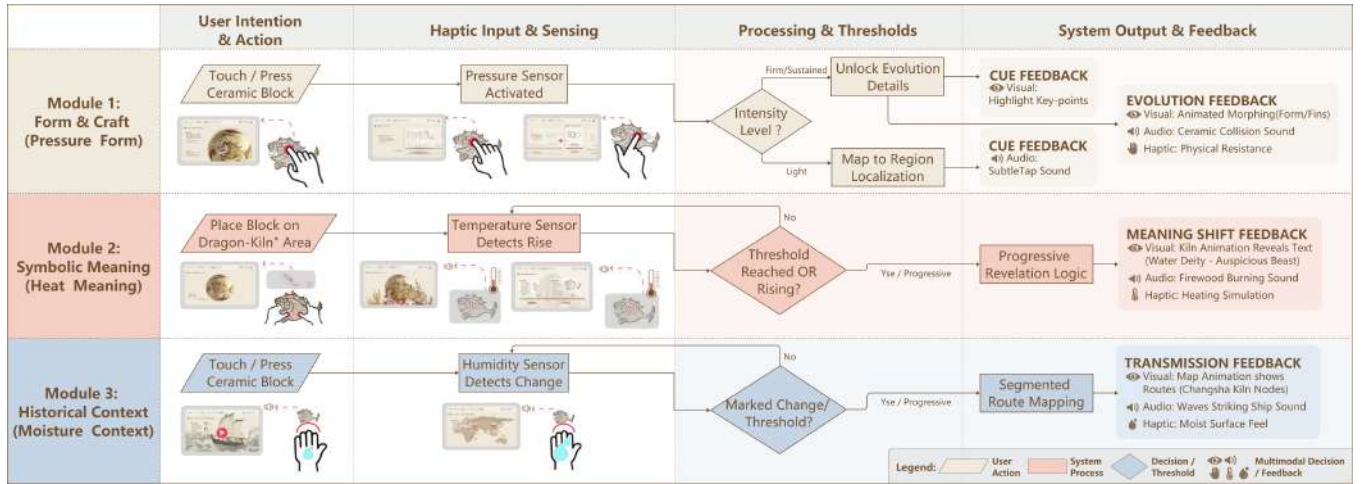


Figure 5: Interaction–Content Mapping for Progressive Interpretation.



Figure 6: Group B interaction process: (a)informed consent, (b)in-situ use, (c)close-up interaction with the tangible relief.

Tactile cues. Heat functions as a sensory bridge into interpretation.

Implementation. Temperature is sensed (Arduino + sensor) and triggers the content when exceeding a threshold (50°C).

Module 3: Historical & Transmission Context (Moisture → Context)

Goal. Situate the motif within Tang maritime exchange and Changsha kiln export circulation, moving from meaning to broader context.

Interaction. Moist touch triggers a “maritime” narrative (Beltung shipwreck), showing animated Maritime Silk Road routes and dissemination to regions such as Southeast Asia, West Asia, and North Africa, paired with sea/ship soundscapes.

Tactile cues. Moisture links bodily sensation to the sea-travel context.

Implementation. Humidity sensing triggers the module/content when exceeding a threshold (60%).

3.4 Interaction Mapping & Walkthrough

Our contribution is a progressive touch-to-learning mapping: actions serve as entry points to interpretive layers, scaffolded by feedback and prompts (R1, R4, R5). As summarized in Figure 5,

users trace the relief to identify form, press to access craft details, heat to reveal symbolic interpretations, and use moist touch to enter historical/transmission context. Prompts drive revisiting, forming a *touch–understand–re-touch* loop (Figure 6). The mapping follows two principles: learning-directed triggers (each cue advances one layer) and guided continuation (prompts/feedback sustain exploration and reduce fragmented stopping points) (R4, R5).

4 EXPERIMENT AND RESULTS

Study Design. We compared a text-and-image baseline (Group A) with the tangible prototype (Group B) to test whether the prototype better supports a “touch → feedback → understand → re-explore” chain and improves learning. We recruited 58 university students and retained 40 valid datasets (Group A: 18; Group B: 22); groups were comparable. Both groups completed a knowledge test (form/meaning/context) and an expression task; Group B also completed a questionnaire and interviews.

Procedure. After consent, participants learned within the same time window: Group A used text-and-image materials; Group B used three modules (*Press → Form*, *Heat → Meaning*, *Moisture → Context*) with progressive disclosure and prompts. Both groups then completed the test and expression task; Group B additionally completed the questionnaire and interviews.

Quantitative Results. Group B scored significantly higher on the knowledge test ($M = 11.14, SD = 0.83$) than Group A ($M = 5.22, SD = 1.48$), $t(38) = -15.95, p < .001$, with the largest gains on fine form-detail and decorative-technique items. Group B also reported higher expression scores and positive experience ratings.

Qualitative Results. Touch anchored form recognition; heat/moisture cues scaffolded meaning/context; prompts drove revisiting; contextual cues strengthened resonance. Group A tended to summarize, while Group B produced chained explanations (form → meaning → context).

5 DISCUSSION

We examine whether the prototype couples Touch–Object–Culture into a *touch* → *feedback* → *understand* → *re-explore* chain. The tangible condition outperformed the baseline on knowledge and expression; interviews suggest tactile localization, contextual cues, and prompted revisiting support progression from form to meaning to context.

Why Touch: Entry point and memory anchor. Touch acted as a learning organizer. Relief exploration plus contextual haptic cues (heat/moisture) helped participants move from sensing to explanation, aligning with experiential learning accounts [12, 16, 20] and supporting affective association [1, 8].

Why an Object: Stable referent and revisit loop. A physical replica provides a stable reference that visitors can return to, linking actions to explanations. Prompts encouraged a *touch–look–re-touch* loop that supports attention, verification, and richer expression in informal learning [23].

Cues as transitions for progressive cultural embedding. Benefits depended less on sensing novelty than on mapping embodied cues to progressive content layers. Heat/moisture functioned as interpretive transitions (craft → meaning → context), pacing gradual disclosure and supporting movement across levels of understanding [12, 23].

Beyond this exhibit. The layered coupling strategy can transfer to other heritage replicas where form features are meaning-bearing (e.g., reliefs, textiles, inscriptions). A reusable procedure is: choose tactile “attention anchors,” define a three-layer trajectory, select transition cues, and add prompts that invite revisiting/comparison.

Museum robustness. Our findings suggest the design can support co-located discussion through tactile/contextual cues, while thresholded sensing plus debounce/cooldown logic helps prevent casual touch from causing unintended activation.

6 Conclusion

In museums, heritage learning is often “visible but not learnable” because conservation restricts touch. We developed a tangible prototype for the Changsha kiln *Mójie* motif that maps pressure, temperature, and moisture cues to three layers—form, meaning, and context. In a controlled comparison, the tangible condition outperformed a text-and-image baseline on learning and engagement. We contribute a reusable approach linking tactile participation, a stable physical referent, and layered interpretation; future work will evaluate long-term retention and transfer.

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